

Frontier Construction Shapes Reflective Prompt Evolution in LLM Negotiation

TEAM 2

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Abstract—Buyer-side LLM negotiation is multi-objective: a buyer prompt must trade off agreement, price, and speed across sellers with different personalities and hidden objectives. We study GEPA-style reflective prompt evolution and show that Pareto frontier construction serves as the supervision-and-evidence interface for the LLM mutator, determining which parent candidates and validation profiles guide rewriting. We formalize three operators, flat, per-instance, and per-axis, along two design axes: the number of evaluation attributes contributing to selection S , and the granularity g at which selection is decomposed into independent instance- or attribute-level signals. A controlled-contrast minibatch replays one (price, objective) scenario against three seller personalities, isolating personality as the sole varying factor visible to the mutator. On unseen-price tests, per-axis lifts buyer score by +2.94 over the AgenticPay base prompt while flat regresses by -2.03 at high seller diversity.

I. INTRODUCTION

Buyer-side LLM negotiation is inherently multi-objective: prompts must trade off agreement quality, price, and efficiency across heterogeneous sellers with hidden objectives. Reflective prompt evolution methods such as GEPA [1] optimize prompts by repeatedly selecting a parent from a Pareto frontier and asking an LLM mutator to revise it on the basis of rollout transcripts. The selected parent determines both which prompt is revised and what evidence the mutator sees during revision. In standard practice, GEPA collapses each validation example into one scalar score, so this design choice has been treated as fixed.

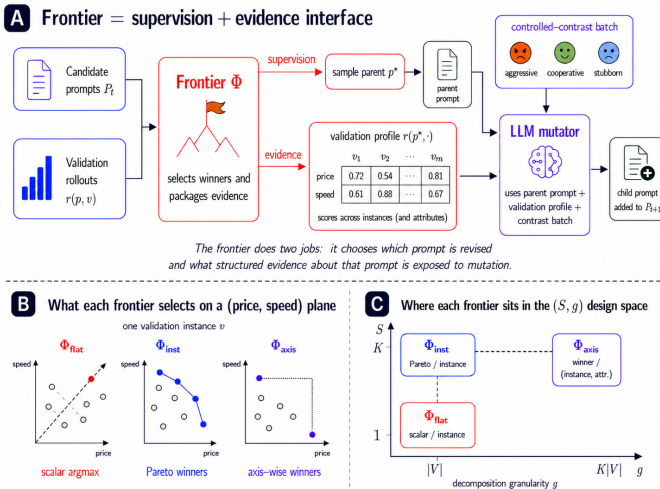


Fig. 1. Frontier design as the supervision-and-evidence interface for reflective prompt evolution. (A) Φ samples a parent p^* (supervision) and its validation profile $r(p^*, \cdot)$ (evidence); both plus a controlled-contrast minibatch reach the LLM mutator. (B) Selection in the (price, speed) plane: scalar argmax / Pareto front / axis winners. (C) Position in the (S, g) space; test-set lifts: +2.10, +2.38, +2.94.

Our project goal is to understand how the construction of the Pareto frontier influences the behavior of the LLM mutator and the generalization of the resulting buyer prompt across heterogeneous sellers. To this end, we treat frontier construction as the design choice in reflective prompt evolution. Three frontier operators, formalized as selection rules on multi-attribute rollout outcomes, are positioned on two axes: supervision dimensionality $S(\Phi)$, counting attributes per instance contributing to selection, and decomposition granularity $g(\Phi)$, measuring separate selection signals the Pareto operator exposes. A controlled-contrast minibatch replays one (price, hidden objective) scenario against three seller personalities, isolating personality as the sole factor visible to the mutator, and two diagnostics (*selection signature*, *mutation signature*) characterize each frontier choice's output.

We make four contributions: (1) reinterpreting Pareto frontier construction as the supervision-and-evidence interface for the LLM mutator rather than a pure ranking step (Section II); (2) formalizing three frontier operators along two design axes, $S(\Phi)$ and $g(\Phi)$ (Section II-0b); (3) introducing a controlled-contrast minibatch and two diagnostics (selection signature, mutation signature) to analyze how frontier structure shapes search dynamics (Section II-0c); and (4) demonstrating that the per-axis operator lifts buyer score by +2.94 while flat regresses by -2.03 under high seller diversity, tracing this gap to specialist preservation and conditional rule formation (Section IV).

II. APPROACH

a) Problem Setup: Buyer-seller negotiation is run in the AgenticPay environment [2]. Sellers vary along three personalities (aggressive, cooperative, stubborn) and six hidden objectives (profit, volume, brand, speed, patient, risk), giving 18 seller types. For each seller type we sample N_p price configurations, with $N_p \in \{1, 3, 5, 7\}$, so a validation set contains $18N_p$ negotiation instances. Personality is observable from the dialogue style and is encoded in the seller system prompt, while the hidden objective enters only through the seller-side reward weights and must be inferred by the buyer from price moves.

The buyer reward S_b on an instance is

$$S_b = \begin{cases} \gamma^{t-1}(D_b + W_b u_b + E_b), & \text{deal,} \\ -F_b(1 - \gamma^{t-1}), & \text{failure,} \end{cases} \quad u_b = \frac{p_{\max} - p}{p_{\max} - p_{\min}}, \quad (1)$$

where D_b, W_b, E_b, F_b are the deal bonus, margin preference, efficiency preference, and failure penalty, γ is a per-round discount, and $p \in [p_{\min}, p_{\max}]$ is the agreed price. Sellers use

the same form with separate weights that encode the hidden objective.

b) *Frontier as Supervision-and-Evidence Interface*: Let $\mathcal{P}_t$ be the candidate pool at generation t and $\mathcal{V}$ the validation set. For each pair (p, v) , a rollout returns $r(p, v) \in \mathbb{R}^K$, where $K = 2$, and r_1 and r_2 are the price and speed scores, both normalized to $[0, 1]$ with higher values better for the buyer. The rollout tensor over $\mathcal{P}_t$ has shape $|\mathcal{P}_t| \times |\mathcal{V}| \times K$. We model frontier construction as a selection operator Φ mapping $\mathcal{P}_t$ to the candidates eligible to be sampled as parents:

$$\Phi_{\text{flat}}(\mathcal{P}_t) = \bigcup_{v \in \mathcal{V}} \arg \max_{p \in \mathcal{P}_t} w^\top r(p, v), \quad (2)$$

$$\Phi_{\text{inst}}(\mathcal{P}_t) = \bigcup_{v \in \mathcal{V}} \text{Pareto}\{r(p, v) : p \in \mathcal{P}_t\}, \quad (3)$$

$$\Phi_{\text{axis}}(\mathcal{P}_t) = \bigcup_{v \in \mathcal{V}} \bigcup_{k=1}^K \arg \max_{p \in \mathcal{P}_t} r_k(p, v). \quad (4)$$

Φ_{flat} collapses each instance to a scalar via AgenticPay weights w . Φ_{inst} retains the full (price, speed) Pareto front per instance. Φ_{axis} keeps only per-axis extremes on every (v, k) pair, admitting candidates that lead on any single attribute of any single instance and thereby creating specialists.

We characterize each operator by two indices. Supervision dimensionality $S(\Phi)$ counts attributes per instance contributing to selection: $S(\Phi_{\text{flat}}) = 1$, $S(\Phi_{\text{inst}}) = S(\Phi_{\text{axis}}) = K$. Decomposition granularity $g(\Phi)$ counts independent selection signals: $g(\Phi_{\text{flat}}) = g(\Phi_{\text{inst}}) = |\mathcal{V}|$ since Φ_{inst} couples the K attributes into one Pareto criterion per instance, while $g(\Phi_{\text{axis}}) = K|\mathcal{V}|$ since it decouples them into independent argmax channels. Beyond selection, Φ also determines the validation profile $r(p, \cdot)$ passed to the mutator, so the same operator controls both the supervision target and the structured evidence guiding rewriting, without explicitly passing seller-objective tags. The default GEPA frontier corresponds to a scalar-score frontier. Φ_{flat} is our scalar baseline, while Φ_{inst} , Φ_{axis} , and the (S, g) design space are our extensions.

c) *Diagnostics and Secret Weapon*: A frontier construction shapes the search in two places: which prompts pass through selection, and what the LLM mutator does with them. We introduce one diagnostic for each and a minibatch design that controls the confounds visible to the mutator.

The *selection signature* ranks available candidates by scalar buyer score at each generation and records the normalized rank of the sampled parent (1 = best, 0 = worst). Averaged over generations, this shows whether Φ tends to sample top, middle, or bottom candidates.

The *mutation signature* is computed on the highest-validation candidate from each run using two prompt-complexity proxies: total character length and the count of conditional markers (if, when, unless, whenever). These proxies do not fully measure strategy quality, but they indicate whether the mutator produces a compact generic prompt or a more conditional prompt encoding personality-specific rules.

Our secret weapon is a controlled-contrast minibatch. At each generation, we fix one (price configuration, hidden seller

objective) pair and replay it against all three seller personalities, producing a 3-instance minibatch. The mutator sees the parent prompt, its validation profile in the (price, speed) plane, and the three rollout transcripts with outcomes. Since price and objective are fixed, personality is the only varying factor, so the minibatch highlights personality-specific failures rather than arbitrary rollout noise.

A higher- S frontier admits unbalanced candidates and a higher- g frontier further preserves per-attribute specialists; combined with the contrast minibatch, these specialists give the mutator personality-specific evidence for conditional rules.

III. IMPLEMENTATION

Episodes are 20-round buyer-seller dialogues in AgenticPay [2], ending either with agreement or at the round cap. The buyer reward follows Eq. (1) with $(D_b, W_b, E_b, F_b) = (10, 80, 10, 15)$ and $\gamma = 0.99$; out-of-range agreements are failures for both sides. Buyer, seller, and mutator are Qwen3-8B, served via sglang during training and OpenRouter at test time, with `enable_thinking=False`, $\tau = 0$, and `seed=42`.

Each generation samples one parent from $\Phi(\mathcal{P}_t)$, draws a 3-instance controlled-contrast minibatch from a 360-instance training pool (3 personalities $\times$ 6 objectives $\times$ 20 prices, with prices disjoint from validation), and queries the mutator for one child. The child is evaluated on the full $18N_p$ validation set. We run all 12 cells, $\{\text{flat, inst, axis}\} \times \{N_p = 1, 3, 5, 7\}$, for 8 generations from the AgenticPay base prompt. The run output is the candidate with the highest validation buyer score among the seed and 8 children. For generalization, each run output and the base prompt are evaluated on a 90-instance held-out test set (18 seller types $\times$ 5 unseen prices) under the same decoding setup.

IV. RESULTS

a) *Result 1: Hierarchical Frontiers Generalize Better and More Stably*: On the held-out test set with unseen prices, all three frontier constructions improve over the AgenticPay base prompt. Table I reports the paired buyer-score lift $\Delta = \text{GEPA} - \text{base}$ on shared instances, broken down by N_p .

TABLE I
BUYER-SCORE LIFT OVER THE AGENTICPAY BASE PROMPT ON THE HELD-OUT TEST SET, WITH MARGINAL AVERAGES. THE BASE MEAN BUYER SCORE IS 7.27.

mode \ N_p	1p	3p	5p	7p	avg
flat	+3.13	+0.90	+6.40	-2.03	+2.10
hier. per-instance	+1.59	+3.18	+1.74	+3.01	+2.38
hier. per-axis	+3.49	+2.22	+3.28	+2.75	+2.94
avg	+2.74	+2.10	+3.81	+1.24	+2.47

Two patterns emerge. First, average lift increases along the design chain $\Phi_{\text{flat}} \rightarrow \Phi_{\text{inst}} \rightarrow \Phi_{\text{axis}}$: +2.10, +2.38, and +2.94, matching the (S, g) ordering. Second, the hierarchical operators are more stable across N_p . Per-axis stays in $[+2.22, +3.49]$ and per-instance in $[+1.59, +3.18]$, while flat ranges from -2.03 at $N_p = 7$ to +6.40 at $N_p = 5$. Because the test prices are not present in the validation set, these gains

cannot be explained by memorization of a particular price range.

b) Result 2: Stable Per-Axis Selection Across Seller Diversity: To understand which candidates pass through each operator, we compute the selection signature defined in Section II-0c. Table II reports the mean parent rank averaged over the 8 generations of each run. Per-axis is the most

TABLE II
SELECTION SIGNATURE: MEAN RANK OF THE SAMPLED PARENT BY SCALAR BUYER SCORE, AVERAGED OVER 8 GENERATIONS. A VALUE OF 1.0 MEANS THE CURRENT BEST CANDIDATE, AND 0.0 MEANS THE CURRENT WORST.

mode	1p	3p	5p	7p
flat	0.59	0.69	0.64	0.25
hier. per-instance	0.45	0.61	0.38	0.51
hier. per-axis	0.49	0.53	0.49	0.53

stable construction: the mean parent rank under Φ_{axis} stays in $[0.49, 0.53]$ across all four values of N_p , near the population median. It neither chases the current scalar winner nor drifts toward weak candidates. Flat is less stable: it is mildly top-biased at $N_p \leq 5$ but drops to 0.25 at $N_p = 7$, where parents are sampled from the bottom quartile of the population at that generation. This behavior follows directly from the operators in Eq. (2) and Eq. (4). Φ_{axis} admits any candidate that wins on one (v, k) channel, so specialists enter the parent pool whenever they are best on one attribute of one instance. Averaging over the $K|\mathcal{V}|$ channels keeps the pool near the population median. In contrast, Φ_{flat} depends on one aggregate scalar per instance. When this scalar becomes noisy under high seller diversity, the parent pool can collapse onto weak candidates.

c) Result 3: Hierarchical Frontiers Produce More Conditional Prompts: Selection rank alone cannot explain the gap: median-rank parents are not inherently better than top-rank ones. We therefore analyze the mutation signature on the best-by-validation candidate of each run.

TABLE III
MUTATION SIGNATURE ON THE BEST-BY-VALIDATION CANDIDATE: PROMPT LENGTH IN CHARACTERS AND CONDITIONAL RULE COUNT.

mode	$N_p = 1$		$N_p = 7$	
	chars	rules	chars	rules
flat	880	7	601	4
hier. per-instance	866	8	1346	11
hier. per-axis	842	7	1039	8

From Table III, at low diversity ($N_p = 1$), three modes produce prompts of similar size. As diversity grows to $N_p = 7$, flat contracts to 601 characters and 4 conditional rules, while per-instance and per-axis grow to 1346/11 and 1039/8 respectively. Under flat selection, the mutator mostly sees generic failures, so it tends to produce a shorter, simpler prompt. Hierarchical modes preserve specialist parents; paired with the contrast minibatch, their failures become personality-specific, giving the mutator evidence for additional conditional rules.

Together, the results suggest one mechanism: higher (S, g) frontiers preserve specialist parents strong on particular at-

tributes such as price or speed (Result 2). Under the controlled-contrast minibatch, these specialists reveal personality-specific failure patterns that the mutator can encode as conditional rules (Result 3), improving generalization across test sellers (Result 1). The flat $N_p = 7$ run shows the reverse pattern: weak parents, the shortest prompt, and a -2.03 test regression.

d) Limitations: Our results use one backbone (Qwen3-8B) and one seed per cell, so the dynamics diagnostics are presented as effect-size evidence. The mutation signature uses a regular expression over four conditional markers and may miss paraphrased conditionals.

V. RELATED WORK

AgenticPay [2] provides the buyer-seller negotiation environment and curated buyer baseline used in this work. Unlike fixed-prompt negotiation agents, we adapt the buyer prompt through reflective evolution.

Our optimization framework builds on GEPA [1], which uses LLM-based mutation and Pareto-based parent selection, but treats each validation example as one scalar score. We instead expose negotiation rollouts as multi-attribute signals and compare flat, per-instance, and per-axis frontier operators.

The per-axis operator is related to Pareto selection [3] and quality-diversity methods such as MAP-Elites [4], since it preserves specialists across attribute-instance cells. Our contribution is to define these cells using negotiation structure and link frontier design to conditional prompt formation and test performance.

VI. CONCLUSION

We showed that frontier construction in GEPA-style reflective prompt evolution shapes which candidates teach the LLM mutator, what evidence it weighs, and how complex the resulting prompts become. Three operators on a rollout tensor were placed in a two-axis design space indexed by S and g ; finer frontiers preserve specialists, expose personality-keyed failures via the contrast minibatch, and produce more conditional prompts that generalize better across seller types. The key finding is that scalarization of the validation outcome is not a neutral default in reflective evolution: it actively hurts at high seller diversity, sampling low-rank parents and producing shorter, less conditional prompts. We achieved the project goal of comparing frontier constructions and identifying the mechanism linking their structure to mutator behavior and generalization, subject to the caveats in the Limitations section. For future work, we would first run multiple seeds to estimate variance and test whether the observed frontier effects are stable, then extend the environment to a multi-buyer, multi-seller market simulation to study how Pareto frontier design should change when prompts must adapt not only to individual sellers but also to evolving market-level dynamics.

TEAM CONTRIBUTION STATEMENT

Chong Zhao and **Tian Xia** designed the frontier operators and the (S, g) design space and built the experiment pipeline. **Manasa Bala** contributed to project framing and design and the experimental pipeline. All wrote the paper.

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